Practical No. 20: * Write a program to manipulate List using key and without key in React

IX. Conclusion

Answer:
Using lists in React helps display multiple items dynamically. Keys are important because they uniquely identify each element. This improves performance and prevents incorrect UI updates when items are added, removed, or changed.

X. Practical Related Questions

Q1. Write a React program to display a list with and without keys

Answer (program1.js)

```
import React from "react";

function Program1() {
  const students = ["Amit", "Riya", "Neha", "Karan"];

  return (
    <div style={{ textAlign: "left" }}>
      <h2>List Without Key</h2>
      <ul>
        {students.map((name) => (
          <li>{name}</li> // No key
        ))}
      </ul>

      <h2>List With Key</h2>
      <ul>
        {students.map((name, index) => (
          <li key={index}>{name}  - {index}</li> // With key
        ))}
      </ul>
    </div>
  );
}

export default Program1;
```

Q2. Write a React program to delete and update list item using key

Answer (program2.js)

```
import React, { useState } from "react";

function Program2() {
  const [students, setStudents] = useState([
    { id: 1, name: "Amit" },
    { id: 2, name: "Riya" },
    { id: 3, name: "Neha" },
  ]);

  const [newName, setNewName] = useState("");

  // Update using input
  const updateItem = (id) => {
    setStudents(
      students.map((s) =>
        s.id === id ? { ...s, name: newName } : s
      )
    );
    setNewName(""); // clear input
  };

  // Delete
  const deleteItem = (id) => {
    setStudents(students.filter((s) => s.id !== id));
  };

  return (
    <div style={{ textAlign: "left" }}>
      <h2>Student List</h2>

      <input
        type="text"
        placeholder="Enter new name"
        value={newName}
        onChange={(e) => setNewName(e.target.value)}
      />

      <ul>
        {students.map((s) => (
          <li key={s.id}>
            {s.name}
            <button onClick={() => updateItem(s.id)}> Update </button>
            <button onClick={() => deleteItem(s.id)}> Delete </button>
          </li>
        ))}
      </ul>
    </div>
  );
}

export default Program2;
```

App.js (Run one program at a time)

```
import React from "react";
import Program1 from "./program1";
import Program2 from "./program2";

function App() {
  return (
    <div>
      {/* Run Program 1 */}
      <Program1 />

      {/* Run Program 2 */}
      {/* <Program2 /> */}
    </div>
  );
}

export default App;
```

XI. Exercise

Q1. Why is using an index as a key sometimes discouraged in dynamic lists?

Answer:
Using index as a key is discouraged because when list items change (add/delete), the index also changes. This can cause incorrect updates and UI problems.

Q2. What happens if two items have the same key?

Answer:
If two items have the same key, React cannot identify them properly. This may cause wrong updates, duplicate rendering, or unexpected behavior.